

ChE 210 Homework Set #9 Due on Friday, 1 November 2019

(1) Murphy 5.40

(2) Murphy 5.41

(3) Air is heated from 25°C to 150°C prior to entering a combustion furnace. The change in specific enthalpy associated with this transition is 3640 J/mol. The flow rate of air at the heater outlet is 1.25 m³/min and the air pressure at this point is 122 kPa absolute.

- (a) Calculate the heat requirement in kW, assuming ideal gas behavior and that the kinetic and potential energy changes from the heater inlet to the outlet are negligible.
- (b) Would the value of $\Delta \hat{E}_k$ [which was neglected in part (a)] be positive or negative, or would you need more information to answer? If you need more information, what would that information be?

(4) Murphy 6.26